CP CAREERPROOF AI

CareerProof AI

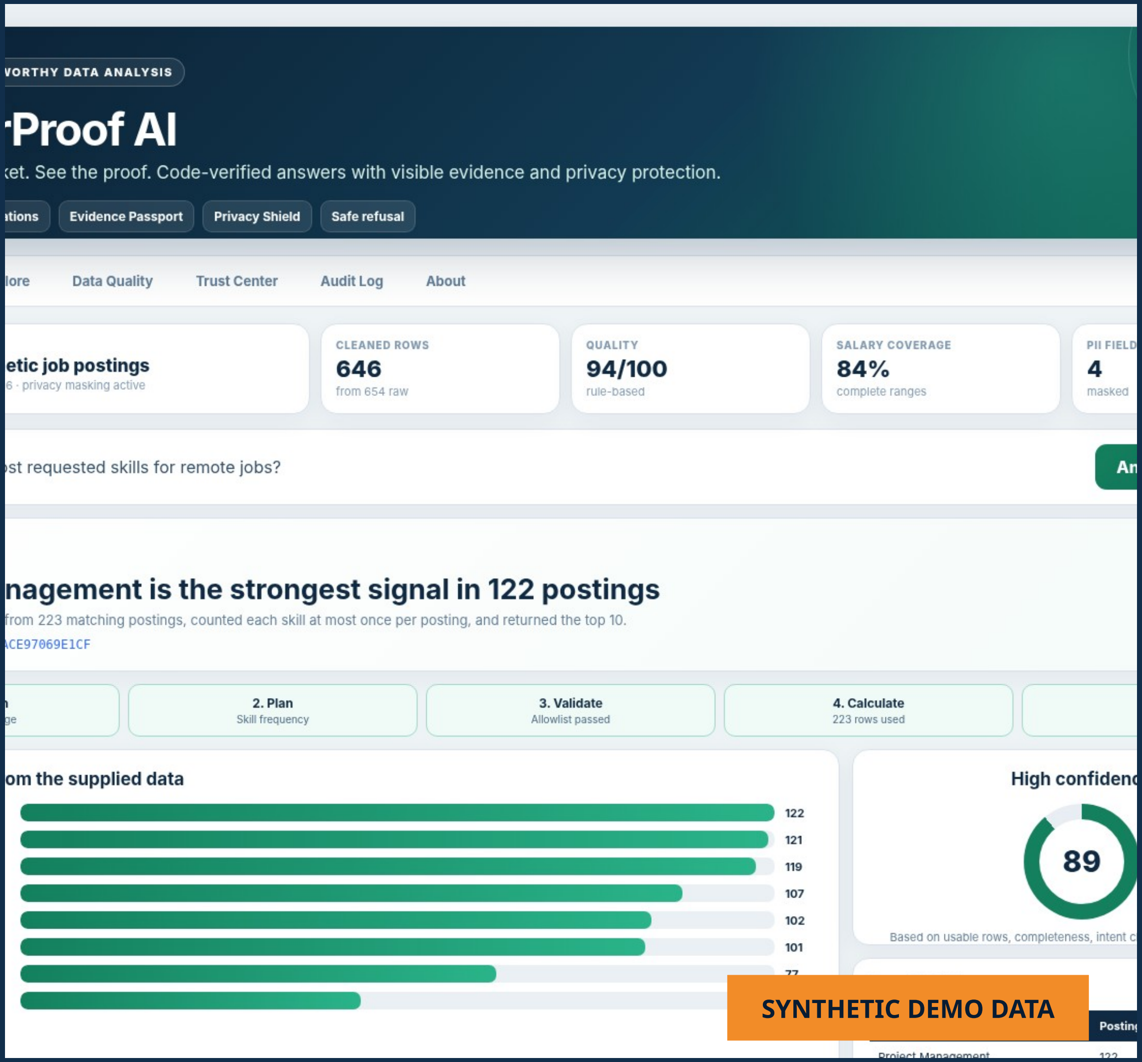
Ask the job market. See the proof.

A trustworthy data-analysis assistant for students and early-career job seekers.

TRACK 2 · TRUSTWORTHY DATA ANALYSIS

AI interprets the question. Code calculates the answer.

Every result includes visible calculations, masked evidence, confidence, and a safe refusal when the dataset cannot answer.



THE PROBLEM

Job seekers have data, not answers

The risk is not only information overload. It is receiving a confident answer with no proof.

Too many listings
Students must compare hundreds of postings across skills, locations, salaries, and experience levels.

Unsupported AI
A general chatbot can produce a plausible number without actually calculating from the uploaded dataset.

Weak decision support
A number without sample size, source rows, or data-quality context can mislead a career decision.

THE TRUST GAP

"Which skills are most common in remote jobs?"

Typical chatbot
Answers in natural language

May invent or blur the calculation

No source table or reproducible plan

CareerProof AI
Interprets the question

Runs an allowlisted Pandas calculation

Shows chart, rows, confidence, and Evidence ID

TARGET USERS High school and college students · recent graduates · career counselors · workforce programs

HOW IT WORKS

A simple flow with a hard trust boundary

Language understanding helps route the question. Only deterministic code is allowed to create factual answers.

1
CSV / XLSX
Bundled or uploaded data

2
Validate + clean
Schema, dates, salaries, duplicates

3
Local AI router
Intent and entity extraction

4
Query plan
Structured and allowlisted

5
Pandas executor
Deterministic calculation

6
Proof output
Chart, table, confidence, audit

TRUST BOUNDARY

The AI never writes or executes Python, SQL, `eval`, or `exec`.

It creates a typed query plan. The validator rejects unsupported columns, operations, filters, and unsafe requests before calculation.

AI side
TF-IDF + logistic regression
Entity matching
Closest supported questions

Code side
Validated filters
Pandas grouping and math
Evidence + confidence rules

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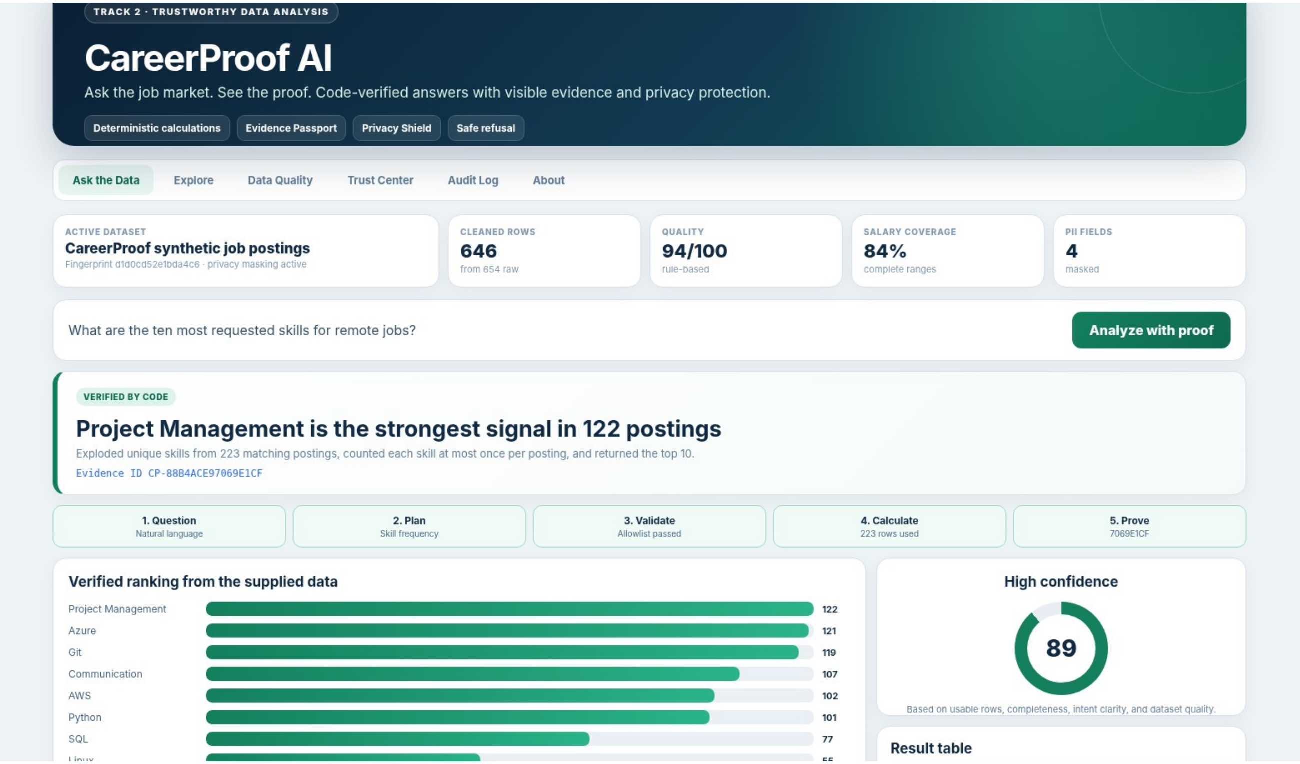
LIVE RESULT

A question becomes a verified answer

What are the ten most requested skills for remote jobs?



HIGH CONFIDENCE

Project Management is the strongest signal in 122 postings

Exploded unique skills from 223 matching postings, counted each skill at most once per posting, and returned the top 10.

223
matching remote postings

89/100
evidence-based confidence

Evidence Passport CP-88B4ACE97069E1CF

The same result can be inspected as a chart, table, source-row preview, query-plan JSON, masked CSV, and portable HTML report.

Demo result is calculated from the bundled synthetic dataset, not generated from model memory.

EVIDENCE PASSPORT

The answer is only the beginning

Each result exposes how it was produced, what data it used, and where uncertainty remains.

TOP SIGNALS IN REMOTE POSTINGS

VISIBLE QUERY PLAN


```
{
  "intent": "skill_frequency",
  "filters": ["work_mode = Remote"],
  "skill_columns": ["required_skills"],
  "limit": 10,
  "chart_type": "bar"
}
```


Calculation

223 rows after filters
Each skill counted once per posting
Sorted by posting count

Confidence

89/100
Driven by row count, completeness, intent certainty, and data-quality score

Privacy

Names, emails, phone numbers, and source IDs are masked before the UI, report, export, or audit log.

Evidence ID CP-88B4ACE97069E1CF changes when the dataset, query plan, or result changes.

BEYOND THE MINIMUM

Four features that make the product memorable

Each feature adds practical value without weakening the trust model.

ID

Evidence Passport

Recompute the ID, then replay the validated plan when the active dataset matches.

?

Refusal Coach

Explains why a question is unsupported and suggests the closest answerable alternatives.

SK

Career Signal Lab

Compares a user skill list with transparent role-level frequency signals. It is not a hiring score.

QL

Cleaning Ledger

Shows every cleaning action, invalid row, missing field, and quality warning instead of hiding them.

Also included

- SCENARIO COMPARE
- MASKED AUDIT LOG
- HTML REPORT EXPORT
- NO API KEY REQUIRED
- CSV / XLSX UPLOAD

Unique does not mean unsafe: every advanced feature still uses the same validated calculation pipeline.

TRUST + QUALITY

Designed to fail safely and prove it

The product is tested against normal questions, malformed data, private-data requests, and prompt-injection attempts.

RISK		VISIBLE CONTROL
Invented numbers	→	**Deterministic Pandas executor**
Arbitrary code	→	**Typed query plan + operation allowlist**
PII exposure	→	**Mask before UI, export, report, or log**
Tiny samples	→	**Minimum group size + confidence downgrade**
Missing fields	→	**Unsupported-question refusal**

62
utomated tests passed

12/12
udge-ready checks passed

94/100
undled data-quality score

0
PI keys required

SAFE REFUSAL DEMO

“Which company has the happiest employees?”

The dataset cannot support that conclusion. The dataset has no employee-satisfaction field.

CP-7AEBDAFC270C6B2F

Known limitations are shown in the product and presentation rather than hidden.

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WHY IT IS SUBMISSION-READY

Built around what the judges actually score

A useful end-to-end product first, with trust, evidence, clear architecture, and a concise story.

Problem + usefulness	25%	Clear early-career user problem
Working prototype	25%	Upload → question → proof → export
Data + AI quality	15%	Local intent AI + deterministic math
Trust + safety	15%	Masking, refusal, confidence, audit
Architecture clarity	10%	Simple input-to-output flow
Demo + storytelling	10%	Success case + limitation case

READY TO SUBMIT

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646 CLEANED ROWS

12 / 12 DEMOS

This dataset is synthetic. Limitations disclosed. No external API required.

Final manual steps Push public GitHub history · record the ≤10-minute video · add team details · upload the verified ZIP